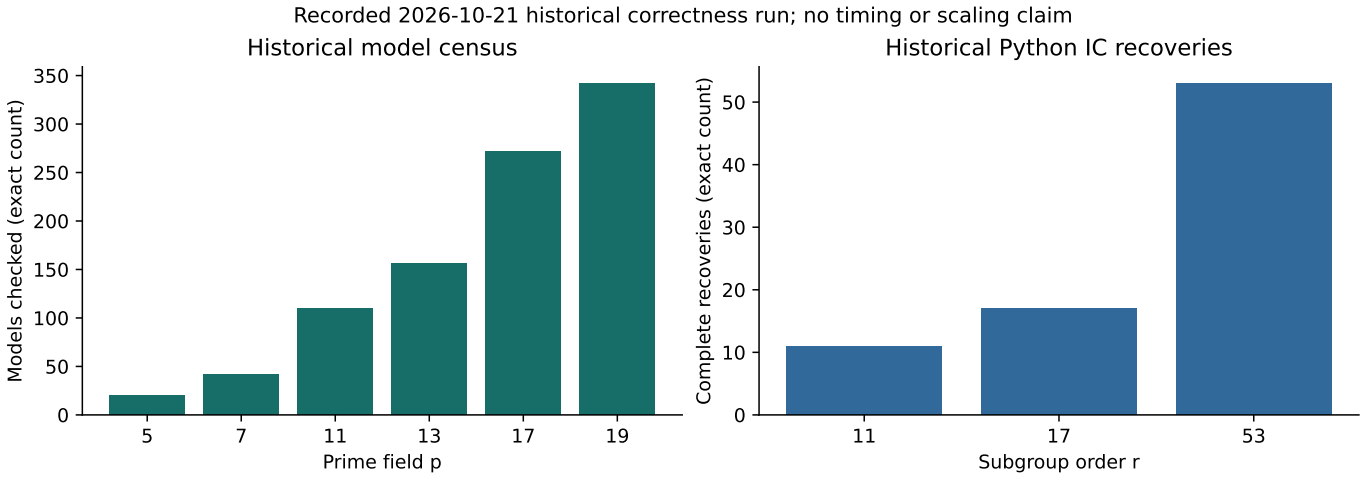
Quadratic-twist evidence import and native audit

Reproducible study record

21 October 2026

Scope. Import the prior experiment records into `aburan28/crypto`. The frozen source and original records remain historical Python evidence. A new C++17 auditor verifies a declared subset independently. This is not a full native port, a performance comparison, or a general ECDLP hardness result.

Historical record. The completed run records 942 nonsingular curve models over primes 5, 7, 11, 13, 17, 19; 4,188,162 ordered S3 triples; 113,342 ordered S4 quadruples over primes 5 and 7; and 81 complete toy IC recoveries after a pairing reduction. These counters are retained as historical observations. The complete census and relation collector were not rerun natively. The input source, transcript, version data, and 12-file manifest are preserved inside `historical/source-and-evidence.tar.xz`.

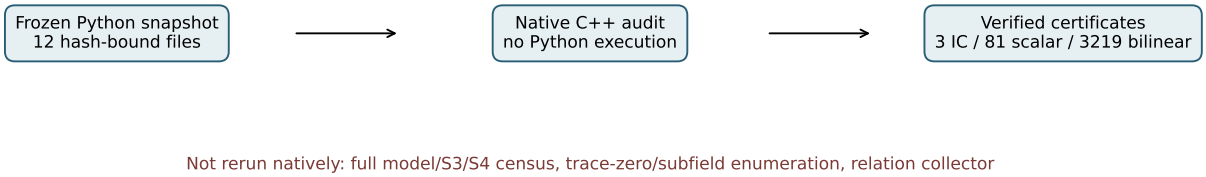


Historical exact finite counts from `historical/results.json`; run started 2026-10-21 05:58:27 UTC. There is no statistical sampling error for these counted domains, and no extrapolation is drawn.

Native audit observations. The final audit at `audits/run-20261021T064417Z/` checked all 12 internal hashes, three saved full-rank IC matrices and their finite-field logarithm certificates, all 81 scalar-pairing identities, all 3,219 bilinearity identities, and the smooth and anomalous twist point-count examples. Miller pairings and elliptic-curve scalar identities were recomputed in C++. This verifies those finite certificates; it does not independently witness the earlier Python process or its full census.

Controls. The native control run accepts the original snapshot, rejects a modified experiment source, and rejects an algebraically false relation even after that result file's checksum is updated. Expected rejections have nonzero exit status. Transcripts are retained beside the native audit receipts.

What the evidence establishes



The archived readable argument distinguishes the plus and minus Frobenius eigenspaces. For a relation $R = A + B$, with R, A rational on the original curve and B a sum of embedded twist-rational points, applying Frobenius gives $R = A - B$, hence $2B = 0$. The odd-primary twist contribution therefore vanishes. The archive also retains a nonzero 2-torsion exception. These are statements with specified hypotheses, not a prohibition of all twist-assisted methods.

The geometric summation-polynomial relation conditions are preserved by the x-fixing quadratic-twist isomorphism over its splitting field. A shared odd prime on a curve and its quadratic twist gives embedding degree two, enabling the specific pairing reductions represented in the saved fixtures. A smooth twist order does not force a smooth original order or a rational homomorphic transfer. The native audit rechecks the two explicit point-count examples.

Requirements still open

The full native census, trace-zero and subfield-descent enumeration, and relation collection are not implemented here. The general written proofs have not been formalized and checked in Lean, and the historical symbolic checks are not registered identity certificates. General GHS construction and general ECDLP hardness remain unestablished.

Conductor, Cargo/Rust, and Lean are absent from this execution environment. The attempted Conductor and Cargo commands exited 127. No coordination reservation, Rust library test pass, or Lean proof check is claimed. These are tooling/implementation gaps, not mathematical obstructions. See `validation/README.md` and `REQUIREMENTS.md`.

Reproduce and audit

```
bash research/quadratic_twist_evidence_20261021/run-audit.sh
bash research/quadratic_twist_evidence_20261021/test-audit.sh
```

Use fresh output directories; the scripts refuse to overwrite prior receipts. The native auditor is `audit.cpp`. Its only library dependency is OpenSSL. The report's Python file only lays out previously recorded values.

No benchmark, leaderboard, registry entry, or canonical IC/rho graph is promoted: this import supplies no admissible cost or speedup row. The scoreboard, leaderboard, progress timeline and lab browser remain unchanged for that reason.

Primary sources preserved by the historical study

- Semaev, summation polynomials: https://eprint.iacr.org/2004/031.
- Gorla–Massierer, trace-zero index calculus: https://arxiv.org/abs/1405.1059.
- Menezes–Okamoto–Vanstone, pairing reduction: https://doi.org/10.1145/103418.103434.
- Gaudry–Hess–Smart, Weil descent: https://www.kantmath.de/publications/papers/weil_descent.pdf.